



JERUSALEM COLLEGE OF ENGINEERING

AN AUTONOMOUS INSTITUTION

Approved by AICTE, New Delhi, Affiliated to Anna University Chennai,
Accredited by NBA, New Delhi and Accredited by NAAC with "A" Grade.
Velachery Main Road, Narayanapuram, Pallikaranai, Chennai, Tamil Nadu - 600100.

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

QUESTION BANK

Subject Code : JAL2501

Subject Name: Machine Learning

Year / Semester: IV / VII

Regulation : 2023

Branch : Common to AI&DS, AI&ML, CS, CSE

Unit No	Unit Name	Prepared by	
		Name	Department
I	Introduction to Machine Learning	Dr.T.Dhanalakshmi	CSE
II	Supervised Learning	Ms.M.Aishwarya	CSE
III	Ensemble Techniques and Unsupervised Learning	Ms.A.Parameshwari	AI&ML
IV	Neural Networks	Ms.A.Anitha	AI&DS
V	Design and Analysis of Machine Learning Experiments	Dr.Lilly Sheba	CS

**Course Instructor /
Course Coordinator**

Scrutinizer

HoD

Dean Academics

Name: Dr.T.Dhanalakshmi
Designation: Associate
Professor

Name: Ms.M.Vanitha
Sheba
Designation: Assistant
Professor

Name: Dr.A.Velmurugan
Designation: Professor

Name: Dr.P.Anuradha
Designation: Professor

Signature:

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Regulation	:	R-2023
Course Name	:	MACHINE LEARNING
Course Code	:	JAL2501
Year / Semester / Branch		IV / VII / AI&DS, AI&ML, CS, CSE
Unit No	:	1
Unit Title	:	INTRODUCTION TO MACHINE LEARNING
No. of Hours	:	9
Weightage in Exam (%)	:	20

PART A

Q.No.	Questions	Marks	BTL	CO
1	Classify the importance of Machine Learning.	2	K2	CO1
2	Identify the purpose of clustering in the K-means algorithm.	2	K1	CO1
3	Mention a few applications of Machine Learning.	2	K1	CO1
4	How Machine Learning Impacts Social Platforms?	2	K2	CO1
5	Differentiate between supervised learning and unsupervised learning.	2	K2	CO1
6	State Null Hypothesis and Alternate Hypothesis.	2	K1	CO1
7	Why is shattering important in Machine Learning?	2	K2	CO1
8	Illustrate the two parameters used in PAC learning.	2	K2	CO1
9	Give examples of inductive bias in Machine Learning algorithms.	2	K1	CO1
10	Mention the role of generalization in Machine Learning.	2	K1	CO1
11	Differentiate between overfitting and underfitting.	2	K2	CO1
12	Classify the two key sources of error that affect a model performance on unseen data.	2	K2	CO1



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PART B

Q.No.		Marks	BTL	CO
1	Apply suitable Machine Learning techniques to three real-world problems and justify the type of learning selected for each.	16	K3	CO1
2	(a) Determine the suitable type of Machine Learning for different real-world scenarios and differentiate their learning processes. (b) Implement vector and matrix operations to represent and process a given dataset in a Machine Learning problem.	8 8	K3	CO1
3	Analyze how VC dimension affects the complexity and generalization ability of a hypothesis class.	16	K4	CO1
4	Summarize how PAC learning uses training-set size, accuracy, and confidence to achieve hypothesis generalization.	16	K2	CO1
5	(a) Compare the bias and variance of two Machine Learning models based on their training and validation performance. (b) Categorize hypotheses and hypothesis spaces based on their effects on model complexity and prediction.	8 8	K4	CO1
6	Illustrate how inductive bias affects hypothesis selection and model generalization with a suitable real-world example.	16	K2	CO1



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Regulation	:	R-2023
Course Name	:	MACHINE LEARNING
Course Code	:	JAL2501
Year / Semester / Branch		IV / VII / AI&DS, AI&ML, CS, CSE
Unit No	:	2
Unit Title	:	SUPERVISED LEARNING
No. of Hours	:	9
Weightage in Exam (%)	:	20

PART A

Q.No.	Questions	Marks	BTL	CO
1	Interpret a suitable dataset that can be used for regression.	2	K2	CO2
2	Give any two examples and applications of supervised learning algorithms.	2	K1	CO2
3	Outline the Simple Linear Regression.	2	K2	CO2
4	Illustrate any two applications of Simple Linear Regression with examples.	2	K2	CO2
5	Differentiate between the Simple and Multiple Linear Regression.	2	K2	CO2
6	Define Least Squares Error Method.	2	K1	CO2
7	Outline the Gradient Descent optimization algorithm.	2	K2	CO2
8	Why is SVM important in Machine Learning	2	K2	CO2
9	Distinguish Stochastic Gradient Descent from Mini-Batch Gradient Descent	2	K2	CO2
10	State the steps involved in the Perceptron learning algorithm.	2	K2	CO2
11	Identify two major generative models.	2	K1	CO2
12	Mention any four applications of Naïve Bayes Classifier.	2	K1	CO2



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PART B

Q.No.	Questions	Marks	BTL	CO																																	
1	Analyze the given data to derive the regression equation and estimate the test score for a student who watches 9 hours of TV. <table><tr><td>Hours, x</td><td>0</td><td>1</td><td>2</td><td>3</td><td>3</td><td>5</td><td>5</td><td>5</td><td>6</td><td>7</td><td>7</td><td>10</td></tr><tr><td>Test score, y</td><td>96</td><td>85</td><td>82</td><td>74</td><td>95</td><td>68</td><td>76</td><td>84</td><td>58</td><td>65</td><td>75</td><td>50</td></tr></table>	Hours, x	0	1	2	3	3	5	5	5	6	7	7	10	Test score, y	96	85	82	74	95	68	76	84	58	65	75	50	16	K4	CO2							
Hours, x	0	1	2	3	3	5	5	5	6	7	7	10																									
Test score, y	96	85	82	74	95	68	76	84	58	65	75	50																									
2	A real estate company wants to predict the selling price (Y in Lakh) of a house based on its size (X_1 in 1000 sq.ft) and number of bedrooms (X_2) using Multiple Linear Regression. The total no. of houses is 10. <table><tr><td>X_1</td><td>1.5</td><td>1.8</td><td>2.4</td><td>1.2</td><td>3.0</td><td>1.6</td><td>2.1</td><td>2.7</td><td>1.4</td><td>2.2</td></tr><tr><td>X_2</td><td>3</td><td>4</td><td>3</td><td>2</td><td>5</td><td>3</td><td>4</td><td>4</td><td>2</td><td>3</td></tr><tr><td>Y</td><td>2.5</td><td>3.0</td><td>3.8</td><td>2.0</td><td>4.8</td><td>2.7</td><td>3.4</td><td>4.3</td><td>2.2</td><td>3.6</td></tr></table> Use the model of the form: $Y=b_0+b_1X_1+b_2X_2$. Where, b_0 is the intercept and b_1, b_2 are the regression coefficients. Predict the house price (Y) for a house with, House Size (X_1) = 2.5 (1000 sq. ft.) and Number of Bedrooms (X_2) = 3	X_1	1.5	1.8	2.4	1.2	3.0	1.6	2.1	2.7	1.4	2.2	X_2	3	4	3	2	5	3	4	4	2	3	Y	2.5	3.0	3.8	2.0	4.8	2.7	3.4	4.3	2.2	3.6	16	K3	CO2
X_1	1.5	1.8	2.4	1.2	3.0	1.6	2.1	2.7	1.4	2.2																											
X_2	3	4	3	2	5	3	4	4	2	3																											
Y	2.5	3.0	3.8	2.0	4.8	2.7	3.4	4.3	2.2	3.6																											
4	(a) Illustrate how Random Forest combines multiple decision trees for classification. (b) Differentiate between generative and discriminative models based on their prediction approaches.	8 8	K2 K2	CO2																																	
5	Interpret how the margin and optimal hyperplane contribute to SVM classification and generalization.	16	K2	CO2																																	



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Analyze the training data to construct a decision tree for predicting Play/Not Play Tennis. Using the tree, classify the tuple $X = (\text{Outlook} = \text{Sunny}, \text{Temp} = \text{Cool}, \text{Humidity} = \text{High}, \text{Wind} = \text{Strong})$ and justify

Day	Outlook	Temp	Humidity	Wind	Play
1	Sunny	Hot	High	Weak	No
2	Sunny	Hot	High	Strong	No
3	Overcast	Hot	High	Weak	Yes
4	Rain	Mild	High	Weak	Yes
5	Rain	Cool	Normal	Weak	Yes
6	Rain	Cool	Normal	Strong	No
7	Overcast	Cool	Normal	Strong	Yes
8	Sunny	Mild	High	Weak	No
9	Sunny	Cool	Normal	Weak	Yes
10	Rain	Mild	Normal	Weak	Yes
11	Sunny	Mild	Normal	Strong	Yes
12	Overcast	Mild	High	Strong	Yes
13	Overcast	Hot	Normal	Weak	Yes
14	Rain	Mild	High	Strong	No

the decision path.

16

K4

CO2



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7

Apply the Naïve Bayes classifier to the given dataset and determine the class of the test instance. $X = (\text{Age} = \text{Youth}, \text{Income} = \text{Medium}, \text{Student} = \text{Yes}, \text{Credit Rating} = \text{Fair})$.

ID	Age	Income	Student	Credit_Rating	Buys_Computer
1	Youth	high	No	Fair	No
2	Youth	high	No	excellent	No
3	middle_aged	high	No	Fair	Yes
4	Senior	medium	No	Fair	Yes
5	Senior	low	yes	Fair	Yes
6	Senior	low	yes	excellent	No
7	middle_aged	low	yes	excellent	Yes
8	Youth	medium	No	Fair	No
9	Youth	low	yes	Fair	Yes
10	Senior	medium	yes	Fair	Yes
11	Youth	medium	yes	excellent	Yes
12	middle_aged	medium	No	excellent	Yes
13	middle_age	high	yes	Fair	Yes
14	Senior	medium	No	excellent	No

16

K3

CO2



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Regulation	:	R-2023
Course Name	:	MACHINE LEARNING
Course Code	:	JAL2501
Year / Semester / Branch		III / VI / AI&DS, AI&ML, CS, CSE
Unit No	:	3
Unit Title	:	NEURAL NETWORKS
No. of Hours	:	9
Weightage in Exam (%)	:	20

PART A

Q.No.	Questions	Marks	BTL	CO
1	List few ensemble techniques.	2	K1	CO3
2	Summarize the algorithm using stacking.	2	K2	CO3
3	Specify the role of Voting in Ensemble Learning.	2	K1	CO3
4	Interpret how bagging improves the stability and accuracy of a model in machine learning.	2	K2	CO3
5	How is the Stacking technique used in ensemble learning?	2	K2	CO3
6	Compare Bagging and Boosting.	2	K2	CO3
7	State Unsupervised Learning with a suitable example.	2	K1	CO3
8	State the concept of clustering in K-means.	2	K1	CO3
9	List two differences between K-means and KNN algorithms.	2	K1	CO3
10	Outline the benefits of K-NN algorithm.	2	K2	CO3
11	Compare Gaussian Mixture Model with K-Means Algorithm.	2	K2	CO3
12	How does the Expectation-Maximization (EM) algorithm work during implementation?	2	K2	CO3



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PART B

Q.No.	Questions	Marks	BTL	CO																																													
1	i) Illustrate the voting mechanism in Ensemble Learning and determine the final prediction for a suitable classification example.	8	K2	CO3																																													
	ii) Outline the working of Stacking in Ensemble Learning and illustrate how multiple base learners are combined for prediction.	8	K2	CO3																																													
2	Analyze Bagging and Boosting techniques and differentiate their approaches to improving model performance.	16	K3	CO3																																													
3	Apply suitable Ensemble Learning techniques to a classification problem and analyze their predictions and performance.	16	K4	CO3																																													
4	Using the K-means algorithm, cluster the following eight points (x, y) into three clusters: $A_1 = (2,10)$, $A_2 = (2, 5)$, $A_3 = (8, 4)$, $B_1 = (5, 8)$, $B_2 = (7, 5)$, $B_3 = (6, 4)$. Initial Cluster Point: $C_1 = (1, 2)$ & $C_2 = (4, 9)$ Use Euclidean Distance Function and show (i) Three cluster centres after the first-round execution (ii) The final three clusters.	16	K3	CO3																																													
5	Summarize the steps of the EM algorithm used for estimating GMM parameters.	16	K2	CO3																																													
6	Consider the given training dataset and use the KNN algorithm to determine the class of the test instance $X = (7.6, 60, 8)$ with $K = 3$. <table border="1" style="margin: 10px auto; width: 80%;"> <thead> <tr> <th>S.No.</th><th>CGPA</th><th>Assessment</th><th>Project Submitted</th><th>Result (Class)</th></tr> </thead> <tbody> <tr><td>1</td><td>9.2</td><td>85</td><td>8</td><td>Pass</td></tr> <tr><td>2</td><td>8</td><td>80</td><td>7</td><td>Pass</td></tr> <tr><td>3</td><td>8.5</td><td>81</td><td>8</td><td>Pass</td></tr> <tr><td>4</td><td>6</td><td>45</td><td>5</td><td>Fail</td></tr> <tr><td>5</td><td>6.5</td><td>50</td><td>4</td><td>Fail</td></tr> <tr><td>6</td><td>8.2</td><td>72</td><td>7</td><td>Pass</td></tr> <tr><td>7</td><td>5.8</td><td>38</td><td>5</td><td>Fail</td></tr> <tr><td>8</td><td>8.9</td><td>91</td><td>9</td><td>Pass</td></tr> </tbody> </table>	S.No.	CGPA	Assessment	Project Submitted	Result (Class)	1	9.2	85	8	Pass	2	8	80	7	Pass	3	8.5	81	8	Pass	4	6	45	5	Fail	5	6.5	50	4	Fail	6	8.2	72	7	Pass	7	5.8	38	5	Fail	8	8.9	91	9	Pass	16	K3	CO3
S.No.	CGPA	Assessment	Project Submitted	Result (Class)																																													
1	9.2	85	8	Pass																																													
2	8	80	7	Pass																																													
3	8.5	81	8	Pass																																													
4	6	45	5	Fail																																													
5	6.5	50	4	Fail																																													
6	8.2	72	7	Pass																																													
7	5.8	38	5	Fail																																													
8	8.9	91	9	Pass																																													



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Course Name	:	MACHINE LEARNING
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Unit No	:	4
Unit Title	:	NEURAL NETWORKS
No. of Hours	:	9
Weightage in Exam (%)	:	20

PART A

Q.No.	Questions	Marks	BTL	CO
1	Illustrate Perceptron model in Machine Learning.	2	K2	CO4
2	List down the advantages of Multi-Layer Perceptron.	2	K1	CO4
3	Give reasons for using an activation function in a neural network and give one example.	2	K2	CO4
4	Classify the main stages involved in training a Neural Network.	2	K2	CO4
5	Outline the objective of Gradient Descent.	2	K2	CO4
6	Summarize the algorithm for stochastic gradient descent.	2	K2	CO4
7	Mention the main steps of back propagation.	2	K1	CO4
8	Identify the types of Neural Networks.	2	K1	CO4
9	Illustrate about unit saturation or vanishing gradient problem. How can it be solved within a neural network?	2	K2	CO4
10	Specify the role of hyperparameter tuning in model training.	2	K1	CO4
11	Differentiate between Randomized Search CV and Grid Search CV.	2	K2	CO4
12	List the types of regularization used in Neural Networks.	2	K1	CO4



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PART B

Q.No.	Questions	Marks	BTL	CO
1	Explain the architecture and operation of Single-Layer Perceptron (SLP) and Multi-Layer Perceptron (MLP), including their advantages and limitations.	16	K2	CO4
2	Determine the parameter updates of Gradient Descent and Stochastic Gradient Descent for a suitable optimization problem.	16	K3	CO4
3	Categorize shallow and deep neural networks based on their architectural features and learning capabilities.	16	K4	CO4
4	Interpret the differences between shallow and deep neural networks in terms of architecture and learning capabilities.	16	K2	CO4
5	(a) Analyze the vanishing gradient problem and its impact on the training of deep neural networks. (b) Contrast the performance of a Machine Learning model before and after hyperparameter tuning.	16	K4	CO4
6	Apply Batch Normalization, Regularization, and Dropout to improve the training and generalization of a neural network.	16	K3	CO4

Regulation

:

R-2023



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Course Name	:	MACHINE LEARNING
Course Code	:	JAL2501
Year / Semester / Branch		IV / VIII / AI&DS, AI&ML, CS, CSE
Unit No	:	5
Unit Title	:	DESIGN AND ANALYSIS OF MACHINE LEARNING EXPERIMENTS
No. of Hours	:	9
Weightage in Exam (%)	:	20

PART A

Q.No.	Questions	Marks	BTL	CO
1	Illustrate the guidelines for Machine Learning experiments.	2	K2	CO5
2	Summarize the working of K-fold Cross Validation.	2	K2	CO5
3	Why Bootstrapping is important?	2	K2	CO5
4	Define a Confusion Matrix and list its components.	2	K1	CO5
5	Specify the meaning of True Positive and True Negative.	2	K2	CO5
6	Differentiate the two types of errors denoted by the Confusion Matrix with suitable examples.	2	K2	CO5
7	State Precision, Recall and F1-score.	2	K1	CO5
8	Give the formula for Specificity in Confusion Matrix.	2	K1	CO5
9	Interpret the concept of the t-test and its formula.	2	K2	CO5
10	Outline the merits and demerits of t-test.	2	K2	CO5
11	Illustrate the use of McNemar's test.	2	K2	CO5
12	Mention the importance of a paired t-test.	2	K1	CO5



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PART B

Q.No.	Questions	Marks	BTL	CO
1	Explain the experimental guidelines for obtaining reliable and reproducible results in Machine Learning.	16	K2	CO5
2	Analyze different Cross-Validation resampling approaches and compare their suitability for evaluating Machine Learning models	16	K4	CO5
3	Apply Bootstrap resampling to a suitable Machine Learning problem and estimate model performance.	16	K3	CO5
4	Categorize different performance measures based on their suitability for various classification problems.	16	K2	CO5
5	Given TP = 100, TN = 200, FP = 50, and FN = 150, calculate and interpret the Accuracy, Precision, Recall, Specificity, and F1-score of the classifier.	16	K3	CO5
6	Compare statistical tests used to evaluate the performance of two classification algorithms.	16	K4	CO5